ONLINE FOOD DELIVERY

SQL Data Analytics Project

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SQL Course Project | 2024

Tool: MySQL 9.0 | Phases: 1–9

1. Introduction

The rapid growth of the online food delivery industry has generated massive volumes of transactional data across customers, restaurants, delivery agents, and orders. Leveraging this data through structured analytical methods is critical for businesses seeking to maintain a competitive edge and improve operational efficiency. This project presents a comprehensive SQL-based data analytics solution built on a simulated Online Food Delivery dataset, designed to extract actionable business insights using MySQL 9.0.

The primary purpose of this project is to demonstrate the practical application of SQL in solving real-world business problems encountered by food delivery platforms. By constructing a normalized relational database and applying structured query analysis across nine phases, this project explores key performance areas including revenue generation, customer behavior, restaurant performance, and delivery efficiency. Each phase builds upon the previous, creating a cohesive analytical pipeline from raw data to business intelligence.

SQL serves as the backbone of this project, employed not only for data retrieval and aggregation but also for performance optimization through indexing and process automation through triggers. The insights derived are directly applicable to strategic decisions such as targeted marketing, resource allocation, and service improvements. This project reflects the role of a Data Analyst working within a food delivery company, tasked with transforming database records into measurable business value.

2. Problem Statement

Business Context

An online food delivery company operates across multiple cities, managing thousands of daily orders, hundreds of restaurant partners, and a fleet of delivery agents. Despite having vast operational data, the company lacks structured analytical insights to support data-driven decision-making.

The company faces several critical business challenges that this project aims to address:

Business Problems Identified

•Inability to identify top-performing restaurants and underperforming outlets across cities

•Lack of visibility into customer ordering patterns and retention metrics

•No automated monitoring of high-value transactions or delayed deliveries

•Inefficient SQL queries causing performance bottlenecks on large datasets

•Revenue leakage due to unmonitored discount policies and negative value entries • No centralized performance dashboard for management reporting

Key Business Questions

•Which restaurants generate the highest revenue and how do they vary by city?

•Who are the most valuable customers and what is their ordering frequency?

•What is the average delivery time per city and which agents perform best?

•How do discount strategies impact overall revenue?

•Which payment methods are most preferred by customers?

•How can we automatically flag high-value orders and delivery delays?

Solving these problems is essential for enabling the company to improve customer satisfaction, optimize delivery operations, grow restaurant partnerships, and maximize profitability across all business units.

3. Business Objectives

As a Data Analyst for the Online Food Delivery company, the following measurable business objectives were defined to guide the analytical scope of this project:

#	Objective	Focus Area

1	Analyze total and city-wise revenue to identify high-revenue markets and growth opportunities	Revenue
2	Identify the top 10 restaurants by revenue and order volume to support partnership decisions	Restaurant Growth
3	Segment customers by order frequency (Frequent / Regular / Occasional) to enable targeted retention strategies	Customer Retention
4	Evaluate delivery agent performance by average delivery time to identify training needs	Operational Efficiency
5	Monitor high-value orders (above ₹1,000) automatically using triggers for financial oversight	Revenue Monitoring
6	Detect delayed deliveries (above 45 minutes) to improve SLA compliance and customer experience	Operational Efficiency
7	Analyze payment method preferences to optimize payment gateway investments	Customer Insights
8	Implement indexes on key query columns to reduce query execution time and improve scalability	Performance Optimization

4. Dataset Overview

The dataset used in this project is a structured relational database designed to simulate the operational data of an online food delivery platform. It contains transactional, customer, restaurant, and logistical records distributed across six interrelated tables.

Dataset Summary

Attribute	Details
Dataset Name	Online Food Delivery Database
Database Tool	MySQL 9.0
Total Tables	6 normalized relational tables
Time Period	January 2024 – March 2024
Total Records (Orders)	18+ transactions
Cities Covered	Mumbai, Delhi, Bangalore, Hyderabad, Pune, Chennai, Kolkata
Data Type	Transactional, Master, Dimensional

Tables Involved

Table Name	Description	Key Columns
customers	Stores customer personal and location details	customer_id, customer_name, city
restaurants	Contains restaurant profiles, cuisine, and ratings	restaurant_id, restaurant_name, cuisine_type, rating
menu_items	Menu catalogue with pricing and availability	item_id, item_name, price, is_available
delivery_agents	Delivery personnel and vehicle information	agent_id, agent_name, vehicle_type, city
orders	Core transactional table with order details	order_id, order_amount, discount, delivery_time_minutes
order_items	Line-level breakdown of items per order	order_item_id, order_id, item_id, quantity

5. ER Diagram Explanation

The Entity-Relationship (ER) diagram defines the structural blueprint of the food delivery database. It illustrates how entities relate to one another through primary and foreign key constraints, ensuring data integrity and enabling efficient JOIN operations.

Entities & Relationships

Entity	Relationship	Cardinality
customers orders	A customer can place many orders	One-to-Many (1:N)
restaurants orders	A restaurant can receive many orders	One-to-Many (1:N)
delivery_agents orders	An agent can handle many orders	One-to-Many (1:N)
orders order_items	Each order can have many line items	One-to-Many (1:N)
menu_items order_items	A menu item can appear in many orders	One-to-Many (1:N)
restaurants menu_items	A restaurant offers many menu items	One-to-Many (1:N)

Key Constraints

•Primary Keys: customer_id, restaurant_id, agent_id, order_id, item_id, order_item_id — all auto-incremented

•Foreign Keys: orders table references customers, restaurants, and delivery_agents ensuring referential integrity

•NOT NULL constraints on critical fields like order_date, order_amount, customer_name

•DEFAULT constraints on discount (0), is_available (TRUE), and order_status ('Delivered')

•UNIQUE constraint on customer email to prevent duplicate registrations

6. SQL Query Explanation

Query: Total Revenue Calculation

```
SELECT SUM(order_amount - discount) AS total_revenue
FROM orders;
```

Aspect	Detail
Business Question	What is the net total revenue generated after applying all discounts?
Table Used	orders
Function Used	SUM() — aggregates all net order values across the entire dataset
Logic Applied	Each order's discount is subtracted from its gross amount before summing
Business Insight	Provides the bottom-line revenue metric used in executive dashboards and financial reporting
Real-world Use	Helps management assess actual earnings versus gross sales to evaluate discount impact

7. Data Analysis & Insights

The following section presents key analytical findings derived from structured SQL queries executed across the food delivery database. These insights are presented as if reported to business management for strategic decision-making.

7.1 Revenue Analysis

Finding

Total net revenue after discounts stands at ₹10,545. Mumbai and Hyderabad are the top revenue-generating cities, together accounting for nearly 45% of total platform revenue.

- UPI is the most preferred payment method, followed by Card and Cash
- High-value orders (above ₹1,000) contribute disproportionately to total revenue

•Discount-heavy orders show lower average order values, suggesting discount optimization is needed

7.2 Customer Behavior

Finding

Customers Aarav Sharma and Rohan Mehta are repeat buyers with 2+ orders each, indicating strong platform loyalty. Occasional customers represent the largest segment but contribute least per head.

•Repeat customers place orders more frequently during evening and night time slots

•Customers from Mumbai and Delhi show the highest average order values

•Registration trend shows steady customer growth from January to October 2023

7.3 Restaurant Performance

Finding

South Spice (Hyderabad) leads with a 4.9 rating and highest order volume. Biryani House follows closely with strong revenue performance. Italian and Indian cuisines dominate platform orders.

•Top 3 restaurants by revenue account for over 40% of total platform revenue

•Restaurants with ratings above 4.5 receive 60% more orders on average

•Chinese and Japanese cuisine restaurants show higher average order values

7.4 Delivery Performance

Finding

Average delivery time across all cities is 37.5 minutes. Hyderabad achieves the fastest average at 19 minutes, while Chennai records the slowest at 60 minutes.

•5 out of 18 orders experienced delays exceeding 45 minutes — all automatically logged via trigger

- Bike delivery agents consistently outperform bicycle agents in delivery speed
- Peak hour orders (evening 6–9 PM) show 20% longer delivery times on average

8. Performance Optimization

As dataset volumes scale, query performance becomes critical. This project implements systematic performance optimization strategies to ensure efficient SQL execution.

Indexing Strategy

Indexes were created on the most frequently queried columns to eliminate full table scans and enable direct row lookup:

```
CREATE INDEX idx_order_date     ON orders(order_date);
CREATE INDEX idx_customer_name   ON customers(customer_name);
CREATE INDEX idx_restaurant_name ON restaurants(restaurant_name);
CREATE INDEX idx_orders_city     ON orders(city);
CREATE INDEX idx_order_amount    ON orders(order_amount);
```

Index	Column	Benefit
idx_order_date	orders.order_date	Speeds up date-range filtering and monthly trend queries
idx_customer_name	customers.customer_name	Faster customer lookup without scanning all rows
idx_restaurant_name	restaurants.restaurant_name	Efficient restaurant search in JOIN operations
idx_orders_city	orders.city	Optimizes city-wise grouping and filtering queries
idx_order_amount	orders.order_amount	Fast retrieval of high-value order queries

Why Indexes Were Not Added on Primary Keys

Primary key columns (customer_id, restaurant_id, agent_id, order_id) are automatically indexed by MySQL. Creating duplicate indexes on these columns would waste storage and slow down write operations without providing additional read benefits.

Query Optimization with EXPLAIN

The EXPLAIN statement was used to verify that MySQL correctly utilizes indexes during execution, confirming that query cost was reduced from full table scans (type: ALL) to index range scans (type: range or ref).

9. Key Findings Summary

#	Key Finding	Business Impact
1	Mumbai and Hyderabad generate the highest revenue	High-priority markets for investment and expansion
2	South Spice (4.9 rating) and Biryani House are top performers	Prioritize for featured placement and promotions
3	UPI is the most preferred payment method (40% of orders)	Invest in UPI payment infrastructure and offers
4	5 orders had delivery times exceeding 45 minutes	SLA breach risk — requires agent performance review
5	Repeat customers make up 30% of users but 60% of revenue	Focus retention strategies on high-value customers
6	High-value orders (>₹1,000) flagged automatically via trigger	Real-time financial monitoring enabled
7	Negative discount entries prevented via trigger	Data integrity maintained without manual intervention
8	5 indexes created reducing average query time significantly	Scalable performance for large production datasets

10. Business Recommendations

Revenue Growth

- Launch city-specific promotional campaigns in Mumbai and Hyderabad to capitalize on existing high-revenue momentum
- Introduce premium membership tiers for repeat customers to increase average order value
- Review and cap discount thresholds above ₹100 to reduce revenue leakage

Customer Retention

- Deploy personalized re-engagement campaigns for occasional customers (less than 2 orders)
- Create loyalty reward programs for frequent customers with 5+ orders
- Use ordering pattern data to send time-based push notifications during peak ordering

hours

Operational Improvement

•Investigate delivery delays in Chennai — consider hiring additional agents or optimizing routes

•Establish SLA alerts for deliveries approaching 40 minutes to enable proactive intervention

•Conduct monthly performance reviews for delivery agents based on average delivery time metrics

Cost Efficiency

•Consolidate restaurant partnerships in low-order cities to reduce operational overhead

•Automate discount validation using database triggers to eliminate manual auditing costs

•Implement query caching for dashboard queries to reduce database load during peak hours

11. Conclusion

This project successfully demonstrates the end-to-end application of SQL in solving real-world business challenges faced by an online food delivery platform. Across nine structured phases, the project covered database design, data insertion, exploratory analysis, advanced business intelligence queries, performance optimization, and process automation — creating a comprehensive analytical solution built entirely within MySQL 9.0.

The analysis uncovered critical insights about revenue distribution, customer behavior, restaurant performance, and delivery efficiency that can directly inform strategic decisions. The implementation of indexes reduced query execution overhead, while triggers provided automated monitoring and data integrity enforcement — capabilities that would scale effectively in a production environment.

This project reflects the core competencies of a professional Data Analyst: the ability to structure data, write optimized queries, interpret results in a business context, and communicate findings to stakeholders. The skills demonstrated here form the foundation of a data-driven approach to business operations in the rapidly evolving food technology sector.

12. Future Scope

While this project establishes a strong analytical foundation, several advanced capabilities can be integrated to further enhance its value:

BI Tool Integration

- Connect MySQL database directly to Power BI or Tableau for real-time interactive dashboards
- Build executive-level scorecards with drill-down capabilities by city, cuisine, and time period

Real-Time Analytics

- Implement MySQL event scheduler to automate daily/weekly summary report generation
- Integrate with Apache Kafka for real-time order stream processing and live delivery tracking

Machine Learning Integration

- Build customer churn prediction models using historical order frequency and recency data
- Develop restaurant recommendation engines based on customer preference patterns
- Apply time-series forecasting for demand prediction and inventory planning

Advanced Database Features

- Implement stored procedures for complex reusable business logic
- Use partitioning on the orders table by date for improved query performance at scale
- Add row-level security and audit logging for enterprise-grade data governance

Cloud & Scalability

- Migrate to cloud MySQL (AWS RDS or Google Cloud SQL) for scalability and high availability
- Implement automated backups, replication, and disaster recovery pipelines

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